

COMPLETE THE ADDRESS AND INSTRUCTION.
Instructions with double asterisks ** are parallel branches.

XIC START_NO_1 START 1 pushbutton.

XIC START_NO_2 START 2 pushbutton.

Rung will go false when the timeout preset has been reached. What is the Tag of the status bit for the timeout?

What is the instruction type which will allow the down solenoid to be energized before the timeout?

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[ ] ** TORQUE_1_SET
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Tag for the internal that are set when TORQUE 1 is reached.

[]	**	TORQUE_2_SET
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Tag for the internal that are set when TORQUE 2 is reached.

[] ** TORQUE_3_SET

Tag for the internal that are set when TORQUE 3 is reached.

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[ ] ** TORQUE_4_SET
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Tag for the internal that are set when TORQUE 4 is reached.

For the above TORQUE 1 thru TORQUE 4, what instruction type will allow the down solenoid to be energized before the torque is reached?

Before the output branches:

DN_LS

An input which senses the nut runners are lowered.

TO.DWELL.DN

Time out status bit.

Each motor is placed on a parallel output branch. What internal Tag is used to turn off the individual in the motor output branch?

_____	Torque_1_Set	Internal that turns off MOTOR 1.
_____	Torque_2_Set	Internal that turns off MOTOR 2.
_____	Torque_3_Set	Internal that turns off MOTOR 3.
_____	Torque_4_Set	Internal that turns off MOTOR 4.

For MOTOR 1, MOTOR 2, MOTOR 3, and MOTOR 4 what instruction type will allow the motor to be energized before the torque is reached?

RUNG 2: TIME OUT (RTO Timer)

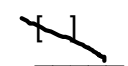
_____ []	DN_LS	An input which senses the nut runners are lowered.
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RUNG 3: TIME OUT ERROR (Latch Output)

_____ []	TO_DWELL_DN	What status bit Tag will be set when the time for tightening the nuts has been exceeded?
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What type of output will remain ON if the rung goes false?

RUNG 4: ERROR RESET (Unlatch Output) and TIME OUT RESET** (Reset Instruction), (Multiple output branches)**

	START_NO_1	START 1 pushbutton have been released.
	START_NO_2	START 2 pushbutton have been released.

What instruction type will be true when the button is released?



RUNG 5: TORQUE 1 SET

What Tag contains the scaled value from the analog torque sensor for Motor 1?

_____ ' AIN_1 _____ A comparative instruction that will turn on the output when the torque value is within the limit.

RUNG 6: TORQUE 2 SET

What Tag contains the scaled value from the analog torque sensor for Motor 2?

_____ AIN_2 _____ A comparative that will turn on the output when the torque value is within the limit.

RUNG 7: TORQUE 3 SET

What Tag contains the scaled value from the analog torque sensor for Motor 3?

_____ AIN_3 _____ A comparative that will turn on the output when the torque value is within the limit.

RUNG 8: TORQUE 4 SET

What Tag contains the scaled value from the analog torque sensor for Motor 4?

_____ AIN_4 _____ A comparative that will turn on the output when the torque value is within the limit.